Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 37)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 1 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 1 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 2 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 2 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 2 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 3 . 1 Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

Waste category	Example / source	Storage or disposal method	
High-level waste			
Intermediate-level waste			
Low-level solid waste			

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0 3 . 2 Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

0 3 . 3 Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

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0 4 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

0 4 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 5 Table 1 shows the rate of heat loss through 1 m^2 of different glazing systems for a 1 °C temperature difference (U-value).

Table 1

Glazing	**Heat loss / W m^{-2} K^{-1}**
Single glazing	5.8
Double glazing, air filled	2.8
Double glazing, argon filled, low-e glass	1.6
Triple glazing, krypton filled, low-e glass	0.8
Double glazing, vacuum (0.2 mm gap)	0.74

0 5 . 1 A house has 20 m^2 of single glazing. The average temperature difference between inside and outside over the heating season is 12 °C. Calculate the reduction in the rate of heat loss if the windows are replaced with argon-filled low-e double glazing.

[3 marks]

Answer ..... W

0 5 . 2 Explain why the gap between the panes should be as large as possible but not too large.

[2 marks]

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0 5 . 3 Explain how low-emissivity glass and inert gas filling each reduce heat loss.

[2 marks]

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0 5 . 4 Explain why vacuum double glazing has a very narrow gap and where it is used.

[2 marks]

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0 5 . 5 State **one** reason why insulation thickness has an optimum value.

[1 mark]

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0 6 Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

Figure 1

Number	Process description
1	Nitrogen gas converted to ammonium ions by Rhizobium in root nodules
2	Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter
3	Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil
4	Amino groups in dead organic matter released as ammonium ions by decomposers
5	Nitrogen and oxygen react during lightning to form NOx
6	Nitrates washed out of soil into rivers

0 6 . 1 Name processes 1 to 6.

[3 marks]

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0 6 . 2 Explain why draining a waterlogged field increases the nitrate available to crops.

[3 marks]

0 6 . 3 Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

[2 marks]

0 6 . 4 Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

[2 marks]

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0 7 Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 7 . 1 Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

0 7 . 2 Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

0 7 . 3 Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

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0 7 . 4 Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

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0 8 . 1 Explain why it is difficult to be certain that a particular flood or heatwave is caused by anthropogenic climate change.

[3 marks]

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0 8 . 2 Explain why historic temperature records collected in towns may overestimate global warming.

[2 marks]

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0 8 . 3 Describe how each of the following sources of proxy data is used to estimate past climates and give one limitation.

[5 marks]

Tree rings (dendrochronology) .....

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Ice cores .....

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0 9 A student is comparing the plant diversity of a grazed and an ungrazed grassland, and estimating the population of grasshoppers in each. In the grazed field 60 grasshoppers were caught, marked and released; three days later 75 were caught, of which 15 were marked.

0 9 . 1 Explain **three** reasons for choosing quadrat positions using random numbers rather than by eye.

[3 marks]

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0 9 . 2 Use the Lincoln index to estimate the grasshopper population of the grazed field.

[2 marks]

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Answer

0 9 . 3 State **three** assumptions that must hold for this estimate to be valid.

[3 marks]

0 9 . 4 Explain how the student should decide the size of quadrat and the number of quadrats to use.

[3 marks]

0 9 . 5 Explain **four** other features of the design that would make the comparison between the two fields more reliable or more valid.

[4 marks]

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1 0 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

1 0 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer ..... times

1 0 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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1 0 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

1 0 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

1 0 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS